

97%

ABERDARE TECHNICAL COLLEGE

ICT TECHNICIAN LEVEL 5

UNIT: COMPUTER NETWORKING AND DESIGN

CONTINUOUS ASSESSMENT TEST (CAT) 3

TIME: 2 HOURS

TOTAL MARKS: 50-60

NAME: Brian Moina

ADMISSION NO: _____

INSTRUCTIONS

1. Answer ALL questions.
2. Show all workings where applicable.
3. Write your answers clearly and neatly.

QUESTION ONE (20 MARKS)

1. Differentiate between the **OSI Model** and the **TCP/IP Model**. (2 Marks)
2. State any two functions of the **Network Layer** in the OSI model. (2 Marks)
3. Define the following terms: (2 Marks)
 - a) IPv4 Address
 - b) IPv6 Address
4. Identify the class of each of the following IP addresses: (2 Marks)
 - a) 10.0.0.1
 - b) 172.16.5.10
5. Differentiate between: (2 Marks)
 - a) Static IP Address
 - b) Dynamic IP Address
6. State any three advantages of IPv6 over IPv4. (3 Marks)
7. Explain the term **CIDR (Classless Inter-Domain Routing)**. (2 Marks)

1] OSI Model

Has 7 layers

Has Reference model
used for learning and
design.

TCP/IP Model

Has 4 layers

It is a Partial
Protocol used on
The internet.

2] Two functions of Net layer in the OSI model.

- Routing → determines the best path for data to travel
- Logical addressing → assigns IP addresses to devices for communication.

3] Define.

a] IPv4 Address → A 32-bit logical address used to identify a device on an IP network. E.g.
192.168.1.1

b] IPv6 Address → A 128-bit logical address designed to replace IPv4 & provide a much larger address space.

4] Identify the Class

a] 10.0.0.1 → Class A

b] 172.16.5.10 → Class B

5] Differentiate.

→ Static IP address

- manually assigned

Does not change unless changed by administrator

Dynamic IP Address

- Automatically assigned by a DHCP server
- Can change whenever the device reconnects to the network

Adv of IPv6 over IPv4

- Provides a much larger space [128-bit]
- Has better security through built-in IPsec support
- Supports automatic address configuration

1] Classless inter-Domain Routing $\rightarrow$ is a method of allocating IP addresses and routing IP ^{packets} ~~addresses~~ using variable length subnet masks instead of fixed address classes.

2] Reduces network traffic by limiting broadcast domains which improves network performance.

3] Improves security and management $\rightarrow$ by dividing a large network into smaller, easier-to-manage subnetworks and controlling access between them.

Router

- Connects different networks together.

- Uses IP addresses to forward data

- Operates at the network layer of the OSI model

Switch

Connects devices within the same network [LAN]

uses MAC addresses to forward data

operates at the data link layer 2 of the OSI model

10 Save the configuration to startup configuration
Connect to the router through console port
Enter global configuration mode and configure basic settings

Cisco 100 Command

11] Enter Global Configuration Mode.

Configure terminal

Configure an IP address on an interface.

Interface GigabitEthernet 0/0

IP Address 192.168.1.1 255.255.255.0

No Shutdown.

Save the Running Configuration.

Copy - running - Config Startup - Config

Explain the Purpose of

a] RIP → A distance-vector routing protocol used to exchange routing information in small networks.

b] OSPF → A link-state routing protocol that finds the shortest path and is suitable for large networks.

c] EIGRP → Advanced routing protocol that provides fast convergence and efficient routing.

d] Improves network security.

Reduces Broadcast traffic.

e] ACL → is a set of rules that permits or denies network traffic. It is used to control access to network resources and improve security.

f] NAT → Maps one private IP address to one public IP address.

PAT → Allows many private IP addresses to share one IP address using different port numbers.

h] Prevents unauthorized devices from connecting.

Improves network security by limiting MAC addresses.

i] STP → Prevents switching loops by blocking redundant paths while keeping back up links available.

18] LACP → Combines multiple physical links to one logical link to increase bandwidth & provide redundancy.

19] - Configure the SSID.

- Set WPA2/WPA3 security & strong password.

- Assign an IP address and configure the wireless channel.

20] DHCP → automatically assigns IP address & other network settings to devices.

DNS → Translates domain names into IP addresses.

21] SSID → is the name of wireless network. It helps users identify and connect to the correct Wi-Fi network.

22] WPA2/WPA3 Encryption with a strong password. Change the default administrator username or password.

23] Network Privilege → permissions granted to the user that determines what resources or actions they can access on a network. E.g. permission to access shared files or install software on a network computer.

~~Ques~~ Ques A

a] SYN → Client sends a SYN packet to the server to request a connection and synchronize sequence numbers.

b] SYN-ACK → Server replies with a SYN-ACK packet to acknowledge the client's SYN and send its own SYN.

c] ACK → Sends an ACK packet to confirm receipt of server's SYN-ACK, completing the connection setup.

d] State the purpose of the TCP 3-way handshake. To establish a reliable TCP connection between the client and server by synchronizing sequence numbers and confirming that both devices are ready to communicate.

25] Wireshark → a network protocol analyzer used to capture and analyze network traffic.

Two uses

- Troubleshooting network problems by identifying errors, delays or packet loss.
- Monitoring and analyzing network traffic to detect security threats to unauthorized activity.

26] Display Filter used.
tcp.

b] Advantage of Packet Analysis

- Packet analysis helps identify the exact cause of network problems such as packet loss, protocol errors or slow communication, making troubleshooting faster and more accurate.

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